Firm and Trade Dynamics: Sunk Costs, Customer Capital, and Buyer–Supplier Search

Daniel Yi Xu (Duke)

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Dynamic Decisions in Trade

- Many trade decisions are **forward looking** and **history dependent**
  - Whether and where to export or import
  - Innovation and quality upgrading
  - Forming and replacing buyer–supplier relationships

  They typically involve sunk, fixed, or convex adjustment costs

- Why model dynamics?
  - **Empirical**: micro data show strong persistence in export status, life-cycle growth after entry, and slow responses to shocks
  - **Macro**: short-run and long-run trade responses to tariffs or exchange rates differ a lot, and so do the welfare implications of the transition
  - **Trade**: identify the deep sources of trade frictions. Static models load everything on fixed versus variable costs (Melitz; Eaton, Kortum and Kramarz); post-entry dynamics tell us what these costs are (Eaton et al.; Alessandria, Arkolakis and Ruhl)

- See also Sam Kortum's "In Search of Trade Frictions"

Roadmap

1. Facts on exporter dynamics: persistence, entry and exit, life cycles (Alessandria, Arkolakis and Ruhl, 2021)
2. The canonical sunk-cost model (Baldwin and Krugman, 1989; Dixit, 1989; Roberts and Tybout, 1997; Das, Roberts and Tybout, 2007)
3. Extending the canonical model
   - Customer capital: Fitzgerald, Haller and Yedid-Levi (2024)
   - Endogenous productivity: Aw, Roberts and Xu (2011)
4. Aggregate implications in general equilibrium (Alessandria and Choi, 2007, 2014; Ruhl, 2008)
5. Trade policy uncertainty and gradual adjustment (Alessandria, Khan, Khederlarian, Ruhl and Steinberg, 2025)
6. Firm-to-firm trade: two-sided search, bargaining and network dynamics (Eaton, Jinkins, Tybout and Xu, forthcoming)

Appendix: continuous-time Bellman equations with Poisson and birth–death processes

Outline

1 **Facts on Exporter Dynamics**

2 The Canonical Sunk-Cost Model

3 Customer Capital and Endogenous Productivity

4 Aggregate Implications

5 Trade Policy Uncertainty and Gradual Adjustment

6 Firm-to-Firm Trade: Two-Sided Search

Fact 1: Export Status Is Persistent

Table 2: The extensive margin

	Export status$_t$			
	(1)	(2)	(3)	(4)
log sales$_t$	0.129***	0.053***	0.053***	0.043***
	(0.007)	(0.003)	(0.003)	(0.004)
exporter$_{t-1}$		0.640***	0.593***	0.636***
		(0.012)	(0.011)	(0.012)
exs$_{t-1}$			0.217***	0.220***
			(0.021)	(0.020)
N	76,662	76,662	76,662	76,662
adj. R^2	0.330	0.618	0.622	0.610

Columns 1–3 include industry and year fixed effects. Column 4 includes year fixed effects. Standard errors, clustered at the industry level, are reported in parenthesis. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Alessandria, Arkolakis and Ruhl (2021), Table 2. Linear probability model of current export status.

- Past export status is the main predictor of current status: coefficient 0.64, and adding it raises the adjusted R^2 from 0.33 to 0.62
- Conditional on past status, current sales matter much less (0.129 falls to 0.053)
- The lagged export–sales ratio also predicts participation (0.22)
- Hard to rationalize with static models in which exporting depends only on current profitability

Fact 2: Entry and Exit

Table 3: Exporter entry and exit

	Stopper_t		Starter_t		
	(1)	(2)	(3)	(4)	(5)
$\log \text{sales}_{t-1}$	0.003		0.027***	0.028***	
	(0.003)		(0.003)	(0.003)	
$\log \text{exports}_{t-1}$	−0.032***	−0.022***			
	(0.003)	(0.001)			
starter_{t-1}	0.244***	0.207***			
	(0.010)	(0.005)			
starter_{t-2}	0.119***	0.084***			
	(0.013)	(0.005)			
$\log \text{destinations}_{t-1}$		−0.075***			0.004***
		(0.002)			(0.0001)
$\log \text{months}_{t-1}$		−0.100***			
		(0.002)			
exporter_{t-2}			0.214***	0.185***	0.158***
			(0.009)	(0.011)	(0.002)
exs_{t-2}				0.211***	
				(0.054)	
Market	World	Country	World	World	Country
N	15,631	324,297	47,289	47,289	20,598,517
adj. R^2	0.157	0.319	0.109	0.111	0.036

Alessandria, Arkolakis and Ruhl (2021), Table 3.

- Exit falls with export size and *tenure*: last year's entrants are 24 percentage points more likely to stop (12 points for entrants two years ago)
- Firms with more destinations and more months shipped are less likely to stop
- Entry rates are low but rise with size
- Past exporters re-enter more readily (+0.21): experience does not fully depreciate after exit

Fact 3: Export Intensity

Table 4: Export intensity

	Export-total-sales ratio$_t$			
	(1)	(2)	(3)	(4)
exporter$_t$	0.216***	0.242***	0.073***	0.240***
	(0.017)	(0.017)	(0.006)	(0.002)
starter$_t$		–0.093***	0.070***	–0.078***
		(0.009)	(0.007)	(0.004)
stopper$_{t+1}$		–0.087***	–0.028***	–0.097***
		(0.009)	(0.004)	(0.004)
starter$_t$,stopper$_{t+1}$		0.063***	0.012	0.045***
		(0.013)	(0.009)	(0.007)
exs$_{t-1}$			0.543***	
			(0.012)	
exs$_{t-2}$			0.190***	
			(0.010)	
stopper$_{t+2}$				–0.040***
				(0.003)
stopper$_{t+3}$				–0.028***
				(0.003)
N	60,568	60,668	60,668	37,072
Adj. R^2	0.358	0.378	0.692	0.381

All specifications include industry and year fixed effects. Standard errors, clustered at the industry level, are reported in parenthesis, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Alessandria, Arkolakis and Ruhl (2021), Table 4.

- Average export intensity of exporters is 0.22 (column 1)
- Starters and stoppers export a much smaller share of their sales (column 2)
- Export intensity is persistent with distributed lags: coefficients 0.54 and 0.19 on the first two lags (column 3)
- Columns 3–4 can be used to trace out an export-intensity *life cycle*

Fact 4: The Exporter Life Cycle

Table 5: Exporter life cycle: The export-sales ratio

	1	2	3	4	5	6	7	Long run
Starter	14.3	15.1	18.3	20.2	21.8	23.1	24.1	28.0
	–7	–6	–5	–4	–3	–2	–1	
Stopper	24.75	23.6	23.6	22.2	21.5	20.6	14.5	

Alessandria, Arkolakis and Ruhl (2021), Table 5. Export–sales ratio (percent) by years since entry and years before exit.

- Starters enter small: export intensity 14.3% in year 1, 24.1% by year 7, against a long-run level of 28%
- Stoppers shrink before they exit: from about 24% seven years before exit to 14.5% in the final year
- Growth in a new market is gradual, through more destinations and growth within each destination

Fact 5: Entrants Matter Slowly for Aggregates

Table 7: Aggregate effects of entry, exit, and growth

	continuation rate				entrants' share		
Window (months)	1	6	12	36	12	36	60
Panel A: Number							
Firm			80	76	20	30	36
Firm, balanced			85	83	15	21	24
Firm*	63	65	59	41	41	54	63
Firm-destination*	54	64	60	46	40	54	62
Panel B: Export value							
Firm			89	91	11	18	21
Firm, balanced			94	98	6	8	7
Firm*	94	98	98	96	2	7	11
Firm-destination*	85	94	94	92	6	13	19

Panel A: *Continuation rate* is the share of exporters that remain exporters across two windows, e.g., 80 percent of firms who exported in a 12-month window export in the next 12-month window. *Entrant's share* is the share of total exporters accounted for by entrants, e.g., 30 percent of exporters are firms that did not export 36 months prior. Panel B: The columns are defined analogously but for export volumes, rather than firm counts. *From the customs transaction-level data.

Alessandria, Arkolakis and Ruhl (2021), Table 7.

- Entrants are many but small: 20% of exporters did not export a year earlier, but they account for only 11% of export value
- After five years, 36% of exporters are new, accounting for 21% of export value
- Continuation is higher when weighted by value (89%) than by count (80%)
- Cumulative entry takes a long time to show up in aggregate export value

Facts: Aggregate Trade

- Long-run responses of trade volumes to trade policy are much larger than short-run responses
  - U.S.–China: short-run tariff elasticity -2.22, long-run -7.93; 90% of the adjustment takes 6 years in an error-correction model and 19 years with local projections (Alessandria et al., 2025)
  - Related literature on trade-policy uncertainty: Handley and Limão (2017)
- Margins of adjustment: new products (Kehoe and Ruhl, 2013), new destinations, new customers (Arkolakis, 2010)
- Responses to exchange-rate fluctuations differ from responses to trade-cost (policy) changes, although the two often move together in large liberalization episodes
  - Persistence matters: firms pay sunk costs only for changes they expect to last (Baldwin and Krugman, 1989)

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Canonical Model: Setup

- Flow profit $\pi(x, z, \xi)$, where $x \in \{0, 1\}$ is export status, z is production efficiency, and ξ is a foreign demand shock
- As in Melitz (2003), foreign demand for firm i is CES:

$$d_{it}(p_{it}) = \xi_{it} \left(\frac{p_{it}\tau_t^*}{P_t^*} \right)^{-\sigma} D_t^*$$

  so the export profit gain $\Delta\pi(z, \xi) \equiv \pi(1, z, \xi) - \pi(0, z)$ is increasing in z and ξ
- The fixed cost of exporting depends on last period's status:

$$f(x_{t-1}) = \begin{cases} f_0 & \text{nonexporter (new entrant)} \\ f_1 & \text{continuing exporter} \end{cases}, \qquad f_0 > f_1$$

  so $f_0 - f_1$ is **sunk**: paid once on entry and lost on exit
- State (x_{t-1}, z_t, ξ_t); each period $V(x_{t-1}, z_t, \xi_t) = \max\left\{V^0(x_{t-1}, z_t, \xi_t),\ V^1(x_{t-1}, z_t, \xi_t)\right\}$

Notation follows Alessandria, Arkolakis and Ruhl (2021): f_0 for nonexporters, f_1 for continuing exporters.

Canonical Model: Bellman Equations

- Values of selling only at home and of exporting:

$$V^0(x_{t-1}, z_t, \xi_t) = \pi(0, z_t) + \beta\, \mathbb{E}\big[V(0, z', \xi') \,\big|\, z_t, \xi_t\big]$$
$$V^1(x_{t-1}, z_t, \xi_t) = \pi(1, z_t, \xi_t) - f(x_{t-1}) + \beta\, \mathbb{E}\big[V(1, z', \xi') \,\big|\, z_t, \xi_t\big]$$

- Export iff $V^1 \geq V^0$. The indifference condition defines a threshold $z_x(\xi)$ for each lagged status x:

$$f_x - \Delta\pi\big(z_x(\xi), \xi\big) = \beta\, \mathbb{E}\big[V(1, z', \xi') - V(0, z', \xi') \,\big|\, z_x(\xi), \xi\big]$$

- Since $f_0 > f_1$: $z_0(\xi) > z_1(\xi)$. Nonexporters need higher efficiency to start than exporters need to continue
- Option value: a firm may keep exporting even when $\Delta\pi < f_1$, because the right-hand side is positive. If z or ξ recovers, it does not have to pay the sunk cost again

A Transparent Special Case: The Hysteresis Band

- No uncertainty: z and ξ are constant, so $\Delta\pi$ is constant over time
- A continuing exporter compares exporting forever with stopping:

$$\frac{\Delta\pi - f_1}{1-\beta} \geq 0 \iff \Delta\pi \geq f_1$$

- A nonexporter compares entering now (then staying) with never entering:

$$\Delta\pi - f_0 + \beta\frac{\Delta\pi - f_1}{1-\beta} \geq 0 \iff \Delta\pi \geq f_1 + (1-\beta)(f_0 - f_1)$$

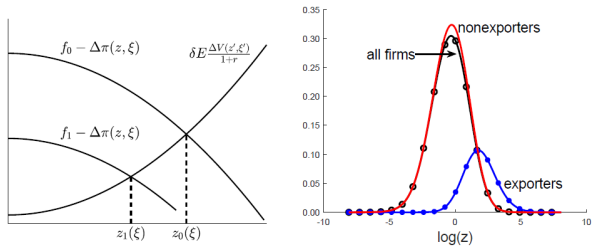
Hysteresis

For $f_1 \leq \Delta\pi < f_1 + (1-\beta)(f_0 - f_1)$, exporters stay and nonexporters stay out: history determines status. The width of the band is the *annuitized* sunk cost $(1-\beta)(f_0 - f_1)$.

- With uncertainty the band widens: the option value of waiting raises the entry threshold and lowers the exit threshold (Dixit, 1989; Baldwin and Krugman, 1989)

Entry and Exit Thresholds

Figure 2: Model properties



(a) Entry and exit thresholds

(b) Firm distributions

Alessandria, Arkolakis and Ruhl (2021), Figure 2. In the figure $\beta = \delta/(1+r)$, with δ the survival probability.

- Nonexporters start above $z_0(\xi)$; exporters stop below $z_1(\xi) < z_0(\xi)$. Exporter and nonexporter productivity distributions overlap

Taking the Model to Data

- Roberts and Tybout (1997): dynamic discrete-choice model of export participation for Colombian plants
  - Sunk costs are identified by state dependence: lagged export status predicts current status *conditional on* observed and unobserved profitability
  - Must separate true state dependence from persistent unobserved heterogeneity (initial-conditions problem)

- Das, Roberts and Tybout (2007): full structural model with heterogeneous export profits and entry costs, estimated by Bayesian MCMC for Colombian chemical producers
  - Entry costs are large and vary a lot across producers
  - Because entry costs are large, expectations about future exporting conditions matter: credible policy changes induce more entry than non-credible ones
  - Most entry and exit is by small exporters, so turnover contributes little to export revenue
  - Subsidies tied to export revenue raise exports more per dollar spent than subsidies that lower entry costs

Successes and Challenges

Successes

- Persistent export participation
- Low entry and exit rates
- Different long-run and short-run responses to shocks
- Overlapping productivity distributions of exporters and nonexporters

Challenges

- To cover the sunk cost, new exporters enter with high productivity. So the model predicts
  - an initial exit hazard that is too low
  - initial export intensity that is too high: it misses the life-cycle pattern (Fact 4)
- Exporting is connected across markets (sequential entry; Albornoz et al., 2012)

- Two ways forward: let demand (customers) be accumulated after entry, or let costs fall with experience

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Market-Specific Customer Capital: Fitzgerald, Haller and Yedid-Levi (2024)

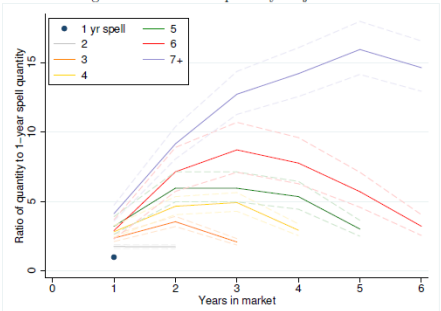
- Life-cycle growth in a new market: falling marginal cost (market-specific learning by doing) or accumulating customers? Use the joint behavior of **quantities and prices**: cost-side stories imply falling prices, demand-side stories need not
- Data: Irish manufacturing census matched to customs records by firm, 8-digit product and destination, 1996–2009
- Specification for a firm–product (i, j) in market k (log quantity or log price):

$$w_t^{ijk} = c_t^{ij} + d_t^{jk} + \beta' \left(s_t^{ijk} \otimes [\, 1/m_i;\ f_k \,] \right) + \varepsilon_t^{ijk}$$

  - c_t^{ij}: firm–product–year effects (marginal cost, quality); d_t^{jk}: product–market–year effects
  - s_t^{ijk}: full set of spell duration $\times$ tenure indicators, plus censoring dummies
  - Permanent heterogeneity: m_i = number of markets the firm ever serves (cost proxy); f_k = number of firms ever serving market k (market-size proxy); spell duration proxies for permanent market-specific demand

Quantity Dynamics

Figure 1: Estimated quantity trajectories



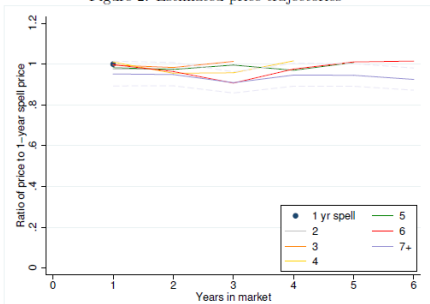
Notes: Figure shows evolution of quantities at the firm-product-market level with market tenure, allowing trajectories to differ by export spell duration. Trajectories are conditional on firm-product-year and market effects. 95% confidence intervals are plotted. Source: CSO and authors' calculations.

Fitzgerald, Haller and Yedid-Levi (2024), NBER WP 21935 version, Figure 1.

- Quantity at entry rises with eventual spell duration
- In spells of 7+ years, quantity roughly quadruples between years 1 and 5
- Quantities fall in the years before exit

Price Dynamics

Figure 2: Estimated price trajectories



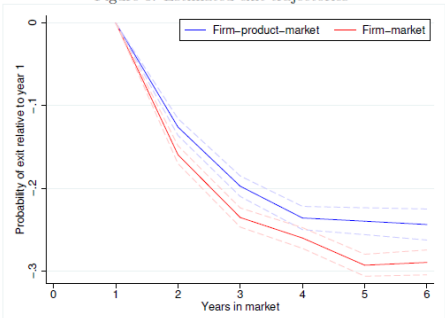
Notes: Figure shows evolution of prices at the firm-product-market level with market tenure, allowing trajectories to differ by export spell duration. Trajectories are conditional on firm-product-year and market effects. 95% confidence interval for spells of 7+ years is plotted. Source: CSO and authors' calculations.

Fitzgerald, Haller and Yedid-Levi (2024), NBER WP 21935 version, Figure 2.

- Prices (markups) at entry are unrelated to spell duration
- No systematic price dynamics with tenure within spells
- Rules out falling marginal cost and "low introductory prices" as the main driver

Exit Hazard

Figure 3: Estimated exit trajectories



Notes: Figure shows reduction in probability of exit at the firm-market and firm-product-market levels with compared to probability of exit in the first year in a market. Trajectories are conditional on firm-year and market and firm-product-year and market effects, respectively. 95% confidence intervals are plotted. Source: CSO and authors' calculations.

Fitzgerald, Haller and Yedid-Levi (2024), NBER WP 21935 version, Figure 3.

- The exit hazard falls steeply with tenure in a market
- Consistent with selection on permanent market-specific demand, plus growing customer capital

Extending the Canonical Model with Customer Capital

- Demand in market k: $Q_t = Y_t^k P_t^{-\theta} D_t^{\alpha} e^{\nu_t}$, with customer base D_t, $\alpha \in (0,1)$, and a demand shock ν_t (permanent + AR(1))
- Customer base starts at $\underline{D}$ on entry and is accumulated through marketing $A_t \geq 0$:

$$D_t = (1 - X_{t-1})\,\underline{D} + X_{t-1}\left[(1-\delta)D_{t-1} + A_{t-1}\right]$$

- Convex investment cost $c(D, A) = A + \phi\, A^2/D$. Pricing stays static: constant markup $P = \frac{\theta}{\theta-1} C$, so

$$\pi_t = \frac{(\theta-1)^{\theta-1}}{\theta^{\theta}}\, Y_t C_t^{1-\theta} D_t^{\alpha} e^{\nu_t} - c(D_t, A_t) - F_t - (1 - X_{t-1}) S_t$$

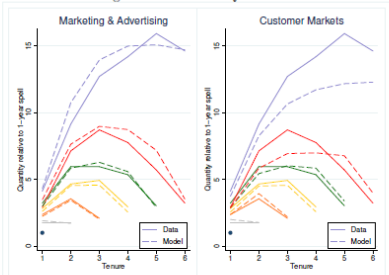
- Optimal marketing for an exporter (FOC, using $\partial D_{t+1}/\partial A_t = 1$):

$$1 + 2\phi\,\frac{A_t}{D_t} = \beta\,\frac{\partial\, \mathbb{E}_t\left[V(1, D_{t+1}, Z_{t+1})\right]}{\partial D_{t+1}}$$

The investment rate rises with the marginal value of customers (a Tobin's q logic). Random fixed and sunk costs ($F \in \{0, F, \infty\}$, $S \in \{0, \infty\}$) give selection on entry and exit; estimated by SMM

Model Fit and What We Learn

Figure 4: Model fit: Quantities



Notes: Figure shows evolution of quantities with tenure, for spells of different duration. Data is from Figure 1. Left panel shows corresponding quantity trajectories for the marketing and advertising model. Right panel shows corresponding quantity trajectories for the customer markets model. All quantities are expressed relative to the quantity in a 1-year spell. Source: CSO and authors' calculations.

- **Marketing & advertising** (left) fits the life-cycle growth in quantities, and hence export-intensity dynamics
- **Customer markets** alternative (right): firms build customers through past sales and lower markups to do so. It fits only with $\theta \approx 32$, i.e. a trade elasticity $\theta/(1-\alpha) \approx 100$
- Implied selling expenses are highest at entry, about 15% of revenue in year 6 of long spells (for $\theta = 3$)

Other Extensions

- How frictions shape export dynamics
  - Adjustment costs in capital and labor; inventories and lumpy shipments
  - Financial frictions: working capital, debt constraints
- Learning about demand: Eaton, Eslava, Jinkins, Krizan and Tybout (2021) combine search for buyers with Bayesian learning about appeal
- The “mirror image”: import dynamics
- Innovation and exporting
  - Exporting raises the return to R&D; R&D raises future productivity and hence export profits
  - Aw, Roberts and Xu (2011): next three slides
  - Many theories of innovation and diffusion, but less evidence with credible identification

R&D and Exporting: Aw, Roberts and Xu (2011)

- Taiwanese electronics plants, 2000–2004: export participation 0.39, R&D participation 0.17; both activities are highly persistent
- Short-run marginal cost $\ln c_{it} = \beta_0 + \beta_k \ln k_{it} + \beta_w \ln w_t - \omega_{it}$; CES demand at home and abroad, with an export demand shock z_{it}, so

$$\ln r^X_{it} = (\eta_X + 1) \ln \tfrac{\eta_X}{\eta_X + 1} + \ln \Phi^X_t + (\eta_X + 1) \ln c_{it} + z_{it}, \qquad \pi^X_{it} = -\tfrac{1}{\eta_X}\, r^X_{it}$$

  (here $\eta_X < -1$ is the demand elasticity)
- Productivity evolves endogenously; the export demand shock is exogenous:

$$\omega_{it} = g(\omega_{it-1}, d_{it-1}, e_{it-1}) + \xi_{it}, \qquad z_{it} = \rho_z z_{it-1} + \mu_{it}$$

  d: R&D (learning by investing); e: exporting (learning by exporting)
- Each period: draw random fixed and sunk costs of exporting, choose e_{it}; then draw R&D costs, choose d_{it}
- Three reasons exporting and R&D move together: selection on (ω, z); R&D raises the value of exporting via g; exporting raises the return to R&D

Estimation Trick: Exponentially Distributed Fixed Costs

- Choice-specific values with lagged status $d_{-1} \in \{0,1\}$ (here for R&D, abstracting from exports): $v^1(s,1) = \beta \mathbb{E}[V(s',1)|s,1]$, $v^1(s,0) = v^1(s,1) - \phi^S$, $v^0(s) = \beta \mathbb{E}[V(s',0)|s,0]$
- Each period the firm draws a fixed cost $\phi \sim$ Exponential with mean σ; it invests iff $v^1 - \phi \geq v^0$, so

$$p^1(s, d_{-1}) = 1 - \exp\big(- [v^1(s,d_{-1}) - v^0(s)]/\sigma \big)$$

- Integrating over ϕ, with $\Delta \equiv v^1 - v^0 > 0$:

$$\mathbb{E}_\phi\big[\max\{v^1 - \phi,\ v^0\}\big] = \int_0^\Delta (v^1 - \phi)\tfrac{1}{\sigma} e^{-\phi/\sigma} d\phi + v^0 e^{-\Delta/\sigma} \ = \ \boxed{v^1 - \sigma\, p^1}$$

- So the value functions are linear in observed choice probabilities:

$$\vec{v}^{\,1}(1) = [I - \beta T_1]^{-1} \beta T_1 \big(\vec{\pi} - \sigma \vec{p}^{\,1}(1)\big)$$

with T_1 the transition matrix of s under investment. Given estimates of $T_0, T_1, \vec{p}^{\,1}$, the costs (σ, ϕ^S) follow from $p^1 = 1 - e^{-\Delta/\sigma}$ (cf. Hotz–Miller)

Aw, Roberts and Xu (2011): Results

Dynamic cost (mil. NT$)	Estimate	S.E.
R&D fixed cost γ_I	67.6	3.9
R&D sunk cost γ_D	354.3	31.4
Export fixed cost γ_F	11.1	0.4
Export sunk cost γ_S	50.8	3.5

Years after shock	2	5	10	15
Δ share exporting (pp)	5.2	9.0	10.0	10.2
Δ share doing R&D (pp)	2.5	3.5	4.1	4.7
Δ mean productivity (%)	0.5	1.5	3.7	5.3

Model 1 estimates and counterfactual from Aw, Roberts and Xu (2011).

- Sunk costs are large relative to fixed costs for both activities
- The model reproduces the joint transition matrix of R&D and export status
- Counterfactual: a permanent increase in export market size raises exporting, then R&D, then productivity, gradually
- With exogenous productivity, the share of exporters rises only 5.5 points after 15 years (vs. 10.2): the innovation channel roughly doubles the long-run export response

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Closing the Model: Alessandria and Choi (2007, 2014), Ruhl (2008)

- Embed the sunk-cost model in a two-country general equilibrium, so prices and wages respond
- Unlike static trade models, add capital and the household's saving decision: transitions take time and use resources
- Composite good aggregates home varieties and imported varieties from foreign exporters:

$$Y_t = \left[\int y_{Ht}(s)^{\frac{\sigma-1}{\sigma}} \mu_t(s)\, ds + \int x^*(s)\, y_{Ft}(s)^{\frac{\sigma-1}{\sigma}} \mu_t^*(s)\, ds \right]^{\frac{\sigma}{\sigma-1}}$$

  where $s = (x_{-1}, z, \xi)$, $\mu_t(s)$ is the measure of firms in state s, and $x^*(s)$ is the foreign firm's export decision

- The composite good is used for consumption, investment and intermediates:

$$Y_t = C_t + \left[K_{t+1} - (1-\delta)K_t\right] + X_t$$

Household, Firms and Equilibrium

- Household chooses consumption and saving:

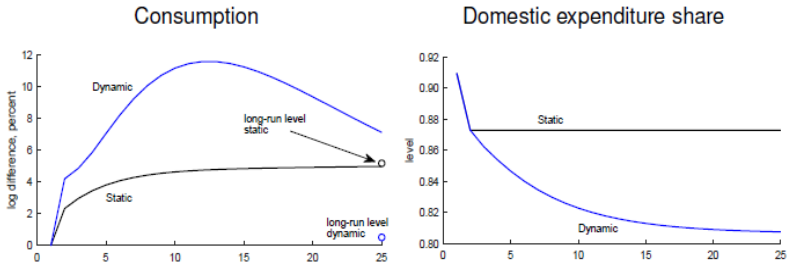
$$\max_{\{C_t, K_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \quad \text{s.t.} \quad C_t + K_{t+1} - (1-\delta)K_t = w_t L + r_t K_t + T_t, \quad K_0 = \bar{K}$$

- Firms: Cobb–Douglas technology with unit cost (intermediate price normalized to one)

$$c(z; w, r) = \frac{\left[w^{1-\alpha} r^{\alpha}\right]^{1-\alpha_x}}{z}$$

- Markets: w_t clears labor, $L \equiv 1 = \int l_t(s)\, \mu_t(s)\, ds$; r_t clears capital, $K_t = \int k_t(s)\, \mu_t(s)\, ds$; the mass of firms is pinned down by free entry each period
- Aggregate state $(\mu_t, \mu_t^*, K_t, K_t^*, w_t, w_t^*, r_t, r_t^*)$: constant in steady state, but transitions require solving for the whole path

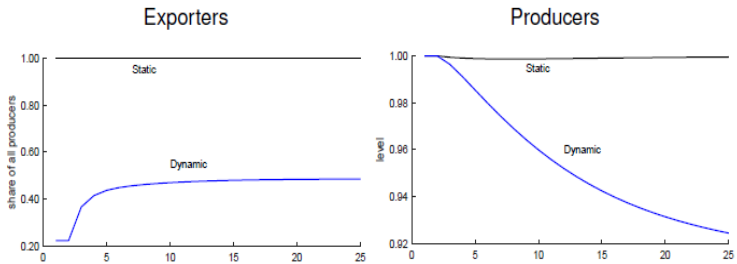
Aggregate Implications of a Tariff Cut: Consumption and Trade



- Imported varieties get cheaper immediately, but exporters enter and grow only gradually: the domestic expenditure share keeps falling for years
- Consumption *overshoots*: resources that would have gone into creating domestic firms are consumed

Alessandria, Arkolakis and Ruhl (2021), Figure 3: dynamic (sunk-cost) vs. static exporter models.

Aggregate Implications of a Tariff Cut: Firms



- The share of exporters rises gradually; the number of producers falls slowly
- Long-run trade elasticity is much larger than in the static model (11.55 vs. 4.00 in the NBER version of AAR)
- Welfare is not well measured by steady-state comparisons: in the dynamic model, the welfare gain (7.8%) is about ten times the change in steady-state consumption (0.75%)

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1 Facts on Exporter Dynamics

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6 Firm-to-Firm Trade: Two-Sided Search

Tariff Uncertainty: Alessandria, Khan, Khederlarian, Ruhl and Steinberg (2025)

- Trade depends on **past**, **current**, and **expected future** trade policy
- The literature typically isolates one of these dimensions, but past policy changes interact with uncertainty about future policy
- U.S.–China trade is an ideal laboratory
  - Dual tariff schedule: Normal Trade Relations (NTR) vs. non-NTR (NNTR, Smoot–Hawley rates)
  - Large cross-good variation in the gap $\tau_g^{NNTR} - \tau_g^{NTR}$
  - U.S. embargo until 1971: a clean zero-initial-trade condition
- Question: how much of the post-2001 surge in Chinese imports reflects WTO accession (resolved uncertainty), and how much reflects slow adjustment to earlier reforms?

Policy Timeline

- 1950–1971: U.S. embargo on China
- 1971: embargo lifted; Chinese goods face high NNTR tariffs
- 1980: China granted NTR, **conditional** on annual renewal by Congress
- 2000–2001: Permanent NTR (PNTR) and WTO accession

- Mean NNTR rate across goods: 29% (s.d. 21%)
- Average applied tariff in 1979: 28% on Chinese goods vs. 6% on goods from NTR countries; by 2001, 2% for both
- So the 1980 reform removed most of the tariff gap, but the *risk* of reverting to NNTR remained until PNTR

Gradual Adjustment and the NTR Gap

- Error-correction model for imports v_{jgt} from country j in good g:

$$\Delta v_{jgt} = \sigma^{SR} \Delta \tau_{jgt} + \gamma \left(v_{jg,t-1} - \sigma^{LR} \tau_{jg,t-1} \right) + \text{fixed effects}$$

  For China: $\sigma^{SR} = -2.22$, $\sigma^{LR} = -7.93$; 90% of adjustment takes 6 years (19 years with local projections)

- "NTR gap" $GAP_g = \log(1 + \tau_g^{NNTR} - \tau_{g,2001}^{NTR})$ and the reduced form

$$v_{jgt} = \sum_{t'=1974}^{2008} \beta_{t'} \, \mathbf{1}(t = t', j = \text{China}) \cdot GAP_g + \delta_{jt} + \delta_{jg} + \delta_{gt} + u_{jgt}$$

- β_t measures exposure to the NTR gap year by year
  - Before 1980, high-gap goods had far lower imports
  - After 1980, the rise is muted by the risk of NTR reversal, and continues long after 2001

Model: Environment

- Many goods g; a continuum of heterogeneous firms in each good
- Firm state: productivity z and iceberg export cost $\xi \in \{\xi_g^L, \xi_g^H, \infty\}$, where $\xi = \infty$ denotes a nonexporter
- Exporting requires a fixed cost: f_0 for nonexporters, $f_1 < f_0$ for continuing exporters
- Two policy regimes $s \in \{1 = \text{NTR},\ 2 = \text{NNTR}\}$ follow a time-varying Markov chain

$$\Pr(s_{t+1} = s' \mid s_t = s) = \omega_t(s, s')$$

- Firms form expectations over future tariffs using ω_t; the sequence $\{\omega_t\}$ is estimated

Model: Static Problem

- Demand facing an exporter of good g under regime s:

$$d_{gt}(p, s) = \left(p\,\tau_{gt}(s)\right)^{-\theta_g} D_{gt}$$

- Profits:

$$\pi_{gt}(z, \xi, s) = \max_{p} \left\{ p\, d_{gt}(p, s) - w \frac{\xi}{z}\, d_{gt}(p, s) \right\}$$

- Optimal price is a constant markup over marginal cost:

$$p^* = \frac{\theta_g}{\theta_g - 1} \frac{w\xi}{z}$$

- Hence export revenue and profit are both proportional to

$$\pi_{gt}(z, \xi, s) \;\propto\; (z/\xi)^{\theta_g - 1}\, \tau_{gt}(s)^{-\theta_g}$$

Model: Dynamic Exporting Problem

- Value of the firm: $V_{gt}(z,\xi,s) = \pi_{gt}(z,\xi,s) + \max\{V^1_{gt}(z,\xi,s),\ V^0_{gt}(z,\xi,s)\}$
- Value of exporting next period, with $f_g(\xi) = f_0$ if $\xi = \infty$ and f_1 otherwise:

$$V^1_{gt}(z,\xi,s) = -f_g(\xi) + \frac{\delta(z)}{1+r}\sum_{s'}\omega_t(s,s')\,\mathbb{E}_{z',\xi'}\left[V_{g,t+1}(z',\xi',s')\right]$$

- Value of not exporting next period:

$$V^0_{gt}(z,\xi,s) = \frac{\delta(z)}{1+r}\sum_{s'}\omega_t(s,s')\,\mathbb{E}_{z'}\left[V_{g,t+1}(z',\infty,s')\right]$$

- Exporting threshold $\bar{z}_{gt}(\xi,s)$ solves

$$f_g(\xi) = \frac{\delta(\bar{z})}{1+r}\sum_{s'}\omega_t(s,s')\left[\mathbb{E}V_{g,t+1}(z',\xi',s') - \mathbb{E}V_{g,t+1}(z',\infty,s')\right]$$

- Expected future tariffs enter through ω_t: a high probability of losing NTR lowers the return to paying f_0 today

Gradual Adjustment Mechanism

- The distribution of firms $\phi_{gt}(z, \xi)$ evolves through
  - productivity transitions $h_g(z', z)$ and exogenous death
  - switching between the iceberg costs ξ_g^L and ξ_g^H
  - the entry and exit decision rules
- Aggregate exports of good g:

$$EX_{gt}(s) = \sum_{\xi \in \{\xi_g^L, \xi_g^H\}} \int p\big(z, \xi, \tau_{gt}(s)\big)\, y\big(z, \xi, \tau_{gt}(s)\big)\, \phi_{gt}(z, \xi)\, dz$$

- A permanent tariff cut raises EX on impact (intensive margin, elasticity θ_g) and then gradually as firms enter and move to ξ^L (extensive margin): the source of $|\sigma^{LR}| > |\sigma^{SR}|$

Calibration Targets

1. The long-run tariff elasticity, -7.93
2. The 1974–2008 sequence of NTR-gap elasticities β_t
3. Chinese firm-level moments by sector, averaged over 2004–2007: export participation rate, exporter exit rate, incumbent premium (exports of incumbents relative to new exporters), and dispersion of log exports

- Persistence of the iceberg cost: $\rho_\xi = 0.91$, identified from the long-run elasticity
- The time-varying switching probabilities $\omega_t(1,2)$ and $\omega_t(2,1)$ are chosen to match the full history of β_t

Estimated Policy Expectations

- Probability of gaining NTR during the 1970s: $\omega(2,1) \approx 0.28$ per year
- Probability of losing NTR, $\omega_t(1,2)$:

$$\omega_t(1,2) \approx \begin{cases} 0.19 & 1980 \\ 0.59 & 1981 \text{ (peak)} \\ \text{falls steadily} & 1986\text{–}1993 \text{ (temporary rise in 1994–96)} \\ 0.05 & \text{late 1990s} \\ 0.011 & 2008 \end{cases}$$

- WTO accession barely changes beliefs: $\omega_t(1,2)$ falls by less than 2 percentage points between 1999 and 2007

Decomposing the NTR-Gap Effects

- Counterfactual without trade-policy uncertainty (policy changes are perceived as permanent)

$$\text{Share of the NTR-gap elasticity } \beta_t \text{ due to TPU} \approx \begin{cases} 61\% & 1980\text{s} \\ 56\% & 1990\text{s} \\ 73\% & 2000\text{s} \end{cases}$$

- Gradual adjustment to the 1971 and 1980 reforms (absent TPU) accounts for about 40% of the effect of PNTR on trade documented by Pierce and Schott (2016)

Takeaways

- Most of the decline in TPU occurred in 1986–1993, not around 2001
- WTO accession played little role in trade-policy expectations
- Post-2001 import surges partly reflect slow adjustment to the 1971 and 1980 tariff changes, which overturns the conventional “WTO shock” interpretation

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Firm-to-Firm Trade: Why We Should Care

- Transaction records between firms are increasingly available
  - Across borders: United States, Latin America, France, …
  - Within countries: Belgium, Japan, Chile, Turkey, Costa Rica, …
- Why trade economists should care
  - International business partners matter for export growth and for supply-chain resilience
  - Trade shocks reach the domestic economy through production networks: intermediates, labor, quality
  - Reassess market inefficiencies in the gains from trade: matching frictions, bilateral market power, and their interaction with other distortions
- Despite a recent surge in the literature, many open questions remain

Endogenous Firm-to-Firm Networks: A Very Incomplete Tour

How firms form relationships matters for (1) rationalizing observed networks and trade flows and (2) predicting responses to policy. Three modeling approaches:

- **Extreme values and discrete choice**
  Oberfield (2018); Eaton, Kortum and Kramarz (2022); Miyauchi (2024); Panigrahi (2022). Dynamics: Lenoir, Martin and Mejean (2023)

- **Fixed costs**
  Bernard, Moxnes and Ulltveit-Moe (2018); Bernard, Dhyne, Magerman, Manova and Moxnes (2022). Dynamics: Dhyne, Kikkawa, Kong, Mogstad and Tintelnot (2023); Lim (2018); Huneeus (2020)

- **Search**
  Bernard, Moxnes and Saito (2019); Demir, Fieler, Xu and Yang (2024); Arkolakis, Huneeus and Miyauchi (2023). Dynamics: Eaton, Eslava, Jinkins, Krizan and Tybout (2021); Eaton, Jinkins, Tybout and Xu (forthcoming)

Challenges

- The individual firm's problem: a large discrete choice that can become combinatorial
- Equilibrium: a large fixed-point problem in which firms' choices are interrelated
- A continuum of partners helps tractability, but is often at odds with the micro data (most firms have few partners) and limits the questions we can ask
  - Example: how market power and prices are determined in firm-to-firm trade
  - Example: how supplier networks readjust under resilience or geopolitical pressure
- Next: a dynamic model with a *discrete* number of partners per firm

Two-Sided Search in International Markets

Eaton, Jinkins, Tybout and Xu (*JPE*, forthcoming)

- A dynamic model of business-to-business trade with
  - two-sided search: foreign suppliers and U.S. buyers (retailers) both choose search effort
  - random matching with a *discrete* number of partners; many-to-many matching
  - forward-looking bargaining over match-specific rents between each buyer and its suppliers
  - multi-variety buyers competing in a monopolistically competitive retail market $\rightarrow$ consumer welfare
- Fit to U.S. customs records on apparel imports, 1996–2011
- Firm life cycles arise endogenously from search frictions and competition, not from exogenous productivity dynamics
- Recovers objects not observed in customs data: the division of match rents, and search effort on both sides

Main Messages

1. **Market access can congest matching.** Greater access of low-quality Chinese suppliers (2005 ATC phaseout and Chinese export-license reforms) *reduced* U.S. consumer welfare
   - High-quality suppliers searched less, lowering the quality at the typical retailer by more than the gain from variety; some small buyers exited
2. **Lower search costs** (an "IT shock", calibrated to 1998–2004 growth in connections) raised consumer welfare substantially, while lowering profits of individual buyers and suppliers through entry on both sides
3. **2018–19 Section 301 tariffs** on Chinese apparel cut consumer welfare on impact, but higher-quality suppliers from other countries quickly formed new matches, offsetting much of the loss

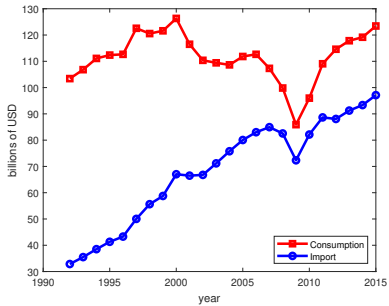
- Buyers and suppliers spend similar amounts on search in aggregate, but buyers spend less per match and capture a larger share of joint surplus

Facts

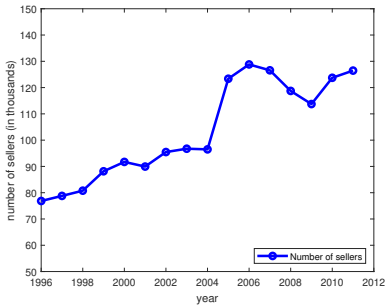
Data

- U.S. Census Bureau Longitudinal Firm Trade Transactions Database (LFTTD), 1996–2011
  - Every import shipment: value, quantity, date, 10-digit HS code
  - Buyer identified by employer identification number (EIN); foreign supplier by a name-and-address string, so buyer–supplier pairs can be tracked over time
- Supplier identifiers are noisy in general (Kamal and Monarch, 2018; Krizan, Tybout, Wang and Zhou, 2020), but relatively accurate for textiles and apparel, where they are used to enforce rules of origin
- Why apparel?
  - Importers are mostly wholesalers or retailers, with nearly separable revenue across product categories
  - Domestic suppliers play a minor role, so customs data cover most of the network
  - Large shocks: the rise of China and the 2005 phaseout of quotas

Aggregate Trends



U.S. apparel consumption and imports

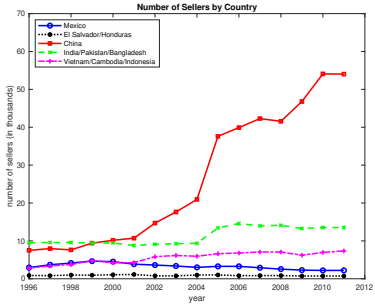


Number of foreign suppliers, 1996–2011

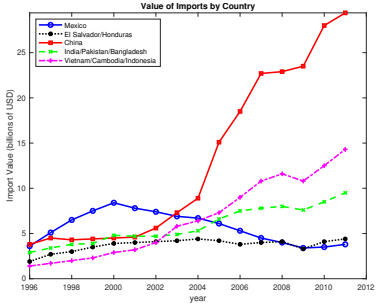
- Import penetration rises from about 30% (1992) to about 80% (2007)
- The number of suppliers grows steadily, with a jump in 2005

Data: WTO, BEA, and LFTTD.

Major Sources of Suppliers: The Rise of China



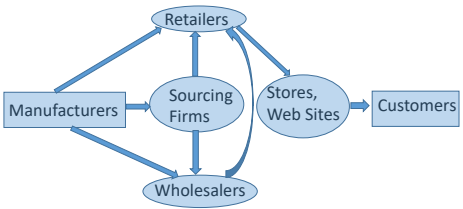
Number of suppliers by country



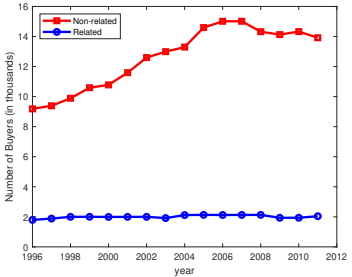
Value of imports by country

- China dominates supplier growth; India, Pakistan, Bangladesh and Vietnam gain after 2005
- Sales per exporter, 1996–2011 average ($000): China 84.2, India 73.4, Indonesia 103.1, Bangladesh 105.1, Vietnam 118.5, Mexico 172.2

Industry Structure: Mostly Arm's-Length Trade



Industry structure



Number of buyers: related party vs. arm's length

- Players: manufacturers, sourcing firms, wholesalers (incl. branded importers), and general merchandise retailers
- Model: three unaffiliated types (suppliers, buyers, consumers). Few importers own factories; affiliated trade is small and arm's-length matches account for virtually all growth

Network Dynamics: Buyers per Supplier

$t \setminus t+1$	0	1	2	3	4	5	6	7	8	9	$\geq$10
1	0.65	0.27	0.05	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.32	0.31	0.21	0.09	0.03	0.02	0.01	0.00	0.00	0.00	0.00
3	0.19	0.22	0.23	0.17	0.09	0.05	0.02	0.01	0.01	0.00	0.01
4	0.13	0.15	0.18	0.18	0.14	0.09	0.05	0.03	0.02	0.01	0.02
5	0.10	0.10	0.13	0.16	0.16	0.12	0.08	0.05	0.03	0.02	0.04
6	0.08	0.07	0.10	0.13	0.14	0.13	0.11	0.08	0.05	0.03	0.07
7	0.07	0.06	0.08	0.09	0.12	0.13	0.12	0.10	0.07	0.05	0.11
8	0.07	0.05	0.05	0.07	0.10	0.11	0.11	0.11	0.09	0.07	0.16
9	0.06	0.05	0.05	0.06	0.08	0.09	0.10	0.10	0.10	0.08	0.24
$\geq$10	0.05	0.03	0.03	0.03	0.04	0.04	0.05	0.05	0.06	0.06	0.56

EJTX, Table 1. Cross-year averages of annual transition rates, LFTTD apparel shipments, 1996–2011. Pairs with a single shipment are not treated as matches.

- Lots of churning: small diagonal terms. Firms with few partners are more likely to lose than gain them

Network Dynamics: Suppliers per Buyer

$t \setminus t+1$	0	1	2	3	4	5	6	7	8	9	$\geq$10
1	0.58	0.26	0.09	0.04	0.02	0.01	0.01	0.00	0.00	0.00	0.01
2	0.34	0.24	0.19	0.10	0.05	0.03	0.02	0.01	0.01	0.00	0.02
3	0.25	0.16	0.18	0.14	0.09	0.06	0.03	0.02	0.02	0.01	0.03
4	0.21	0.11	0.14	0.14	0.13	0.08	0.06	0.04	0.03	0.02	0.06
5	0.19	0.07	0.10	0.12	0.12	0.11	0.07	0.06	0.04	0.03	0.09
6	0.17	0.06	0.08	0.09	0.11	0.11	0.09	0.07	0.05	0.04	0.13
7	0.16	0.05	0.05	0.07	0.09	0.10	0.09	0.09	0.06	0.06	0.19
8	0.15	0.04	0.05	0.06	0.07	0.08	0.08	0.08	0.07	0.06	0.25
9	0.15	0.03	0.03	0.04	0.06	0.07	0.08	0.08	0.07	0.07	0.32
$\geq$10	0.12	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.03	0.03	0.71

EJTX, Table 2. Same construction as Table 1.

- Matches are short-lived: the annual match separation hazard is 0.77, and match survival is well approximated by a Poisson process ($R^2 = 0.98$)
- Even firms with many partners can drop to zero, mainly through exit

Degree Distributions and Sales Heterogeneity

x	Buyers with $\leq x$ suppliers	Suppliers with $\leq x$ buyers
1	0.407	0.798
2	0.554	0.911
3	0.645	0.951
4	0.709	0.970
5	0.743	0.980
6	0.780	0.987
7	0.808	0.991
8	0.823	0.993
9	0.837	0.995
10	0.855	0.996

EJTX, Table 3 (2011).

- Most firms have fewer than 10 foreign partners (86% of buyers, over 99% of suppliers), but both distributions are close to a power law: a few firms are very large
- Suppliers' counts only include their U.S. partners
- The Pareto shape for buyers per supplier rises from 1.99 (2000) to 2.84 (2011), as suppliers per buyer increase
- Next slide: within-buyer heterogeneity across suppliers. Buyers typically have a dominant supplier whose share falls only modestly as the number of suppliers grows

Supplier Heterogeneity within Buyer

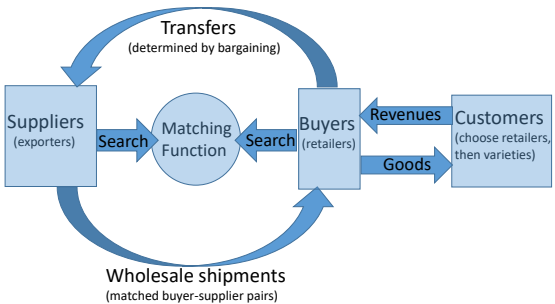
No. suppliers	Mean log imports	Supplier shares in buyer purchases, ordered by supplier size									
		1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
1	0.000	1.000									
2	1.134	0.771	0.229								
3	1.604	0.668	0.240	0.092							
4	1.764	0.608	0.232	0.111	0.049						
5	1.904	0.544	0.231	0.126	0.067	0.032					
6	2.054	0.540	0.218	0.115	0.070	0.039	0.019				
7	2.214	0.502	0.214	0.123	0.074	0.045	0.027	0.019			
8	2.094	0.460	0.212	0.125	0.080	0.054	0.035	0.023	0.011		
9	2.364	0.451	0.201	0.121	0.083	0.055	0.038	0.026	0.017	0.017	
10	2.324	0.420	0.197	0.125	0.084	0.060	0.042	0.029	0.021	0.013	0.007

EJTX, Table 4. Column 2: mean log imports per supplier, relative to buyers with one supplier.

- Buyers with two suppliers spend 113% more per supplier than single-supplier buyers: identifies the dispersion of buyer types
- The within-buyer shares identify the dispersion of supplier types

Model

Model Overview



- Continuous time; three types of agents: foreign suppliers, U.S. buyers (retailers), and a representative consumer
- **Retail market**: no frictions. The consumer values each buyer for its current product set and its amenities
- **Wholesale market**: suppliers and buyers search for each other; each match gives the buyer a variety tailored to it
- Buyers and suppliers continuously choose (i) retail prices, (ii) the division of surplus through bargaining, (iii) search effort

Retail Market: Preferences and Demand

- The consumer spends a fixed amount E on apparel, allocated across a continuum of buyers y:

$$U = \left[\int_y (\mu_y Q_y)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad Q_y = \left[\sum_{x \in J_y} (\xi_x q_{xy})^{\frac{\alpha-1}{\alpha}} \right]^{\frac{\alpha}{\alpha-1}}$$

- J_y: discrete set of products offered by buyer y, one per matched supplier x; μ_y: buyer appeal (amenities); ξ_x: supplier appeal
- Demand for product x at buyer y:

$$q_{xy} = \xi_x^{\alpha-1} \mu_y^{\eta-1} \left(\frac{p_{xy}}{P_y} \right)^{-\alpha} \left(\frac{P_y}{P} \right)^{-\eta} \frac{E}{P}$$

  with $P_y = \left[\sum_{x \in J_y} (p_{xy}/\xi_x)^{1-\alpha} \right]^{\frac{1}{1-\alpha}}$ and $P = \left[\int_y (P_y/\mu_y)^{1-\eta} \right]^{\frac{1}{1-\eta}}$

- The buyer and supplier jointly incur a unit cost c_x; quality-adjusted cost $\tilde{c}_x = c_x/\xi_x$

Pricing: A Multi-Product Retailer Charges a Constant Markup

- Bargaining (below) leads each buyer and its suppliers to set prices that maximize joint retail profit $\pi_y^T = \sum_{x \in J_y} (p_{xy} - c_x) q_{xy}$:

$$q_{xy} + \sum_{x' \in J_y} \frac{\partial q_{x'y}}{\partial p_{xy}} (p_{x'y} - c_{x'}) = 0, \qquad \forall x \in J_y$$

- Each buyer is measure zero, so P is fixed. With within-buyer share $h_x \equiv (p_{xy}/\xi_x)^{1-\alpha}/P_y^{1-\alpha}$: $\frac{\partial \ln q_{x'y}}{\partial \ln p_{xy}} = -\alpha \, \mathbf{1}_{x'=x} + (\alpha - \eta) \, h_x$
- Multiply the FOC by p_{xy}; with $r_x = p_{xy} q_{xy}$, $R_y = \sum_x r_x$, and Lerner index $m_x = (p_{xy} - c_x)/p_{xy}$:

$$r_x (1 - \alpha m_x) + (\alpha - \eta) \, h_x \sum_{x'} m_{x'} r_{x'} = 0$$

- Try $m_x = m$ for all x and use $h_x = r_x / R_y$: $r_x \big[1 - \alpha m + (\alpha - \eta) m\big] = 0$, so $\boxed{(p_{xy} - c_x)/p_{xy} = 1/\eta}$
- The markup is set by competition *across* buyers (η), not across varieties within a buyer (α)

Total Flow Surplus with Discrete Types

- With markup $\frac{\eta}{\eta-1}$, joint profit is a fixed share $1/\eta$ of the buyer's revenue $P_y Q_y = E\left(P_y/\mu_y\right)^{1-\eta}/P^{1-\eta}$:

$$\pi_y^T = \frac{1}{\eta}\frac{E}{P^{1-\eta}}\left[\sum_{x\in J_y}\left(\frac{\eta}{\eta-1}\right)^{1-\alpha}\tilde{c}_x^{1-\alpha}\right]^{\frac{\eta-1}{\alpha-1}}\mu_y^{\eta-1}$$

- Finite types: buyers $i \in \{1,\ldots,I\}$ with appeal μ_i; suppliers $j \in \{1,\ldots,J\}$ with $\tilde{c}_j$. A type-i buyer's supplier portfolio is $\mathbf{s} = [s_1,\ldots,s_J]$ (counts by type); a supplier's buyer portfolio is $\mathbf{b} = [b_1,\ldots,b_I]$:

$$\pi_i^T(\mathbf{s}) = \frac{E}{\eta P^{1-\eta}}\left[\sum_{j=1}^{J}\left(\frac{\eta}{\eta-1}\right)^{1-\alpha}s_j\,\tilde{c}_j^{1-\alpha}\right]^{\frac{\eta-1}{\alpha-1}}\mu_i^{\eta-1}$$

- If $\alpha > \eta > 1$, the exponent $\frac{\eta-1}{\alpha-1} < 1$: diminishing returns to additional suppliers, which makes a buyer's negotiations with its suppliers interdependent. Suppliers with high $\tilde{c}_j$ are "low quality"

Search Costs and Match Separation

- An agent with n^A partners choosing search effort σ meets a new partner at rate $\sigma\theta^A$ and pays flow cost

$$k^A(\sigma, n^A) = \frac{k_0\,\sigma^2}{(n^A+1)^{\gamma^A}}, \qquad A \in \{B, S\}, \quad n^B(\mathbf{s}) = \textstyle\sum_j s_j, \;\; n^S(\mathbf{b}) = \textstyle\sum_i b_i$$

  - $\gamma^A > 0$: a **visibility effect**; firms with many partners find new ones more cheaply (generates fat tails, as in Eaton et al., 2021)
  - $\gamma^A < 0$: the market is "fished out" (generalizes Arkolakis, 2010)

- With quadratic cost, the FOC $\partial k^A/\partial\sigma = \theta^A \times (\text{expected gain from a match}) \equiv \theta^A G$ gives

$$\sigma^* = \frac{(n^A+1)^{\gamma^A}\,\theta^A\,G}{2k_0}$$

  so effort rises with visibility, market tightness, and the value of a match

- Matches end exogenously: incompatibility at rate δ; buyer exit at rate δ^B; supplier exit at rate δ^S

Buyer's Problem

- A type-i buyer with portfolio $\mathbf{s}$ keeps $\pi_i^T(\mathbf{s}) - \sum_j s_j \tau_{ji}(\mathbf{s})$, where $\tau_{ji}(\mathbf{s})$ is the bargained transfer to each type-j supplier. Its value solves the HJB equation

$$(\rho + \delta^B) V_i^B(\mathbf{s}) = \pi_i^T(\mathbf{s}) - \sum_{j=1}^{J} s_j \tau_{ji}(\mathbf{s}) + \max_{\sigma} \left\{ \sigma\, \theta^B \sum_{j=1}^{J} \upsilon_j^S \left[V_i^B(\mathbf{s} + \mathbf{1}_j) - V_i^B(\mathbf{s}) \right] - k^B(\sigma, n^B) \right\}$$
$$+ (\delta + \delta^S) \sum_{j=1}^{J} s_j \left[V_i^B(\mathbf{s} - \mathbf{1}_j) - V_i^B(\mathbf{s}) \right]$$

- υ_j^S: probability that the next supplier met is type j; ρ: discount rate
- Optimal search:

$$\frac{\partial k^B(\sigma_i^B(\mathbf{s}), n^B)}{\partial \sigma} = \theta^B \sum_{j=1}^{J} \upsilon_j^S \left[V_i^B(\mathbf{s} + \mathbf{1}_j) - V_i^B(\mathbf{s}) \right]$$

- Events: lose a type-j supplier at rate $s_j(\delta + \delta^S)$; gain one at rate $\sigma_i^B \theta^B \upsilon_j^S$; exit at rate δ^B

Supplier's Problem

- Constant marginal cost and distinct products for each buyer, so a supplier's matches are separable: the return to search depends only on the expected value of a new match, V_j^S

$$\frac{\partial k^S(\sigma_j^S(n^S), n^S)}{\partial \sigma} = \theta^S V_j^S, \qquad V_j^S = \sum_i \sum_{\mathbf{s}\in\mathbb{S}} \upsilon_i^B(\mathbf{s})\, V_{ji}^S(\mathbf{s})$$

  with $\upsilon_i^B(\mathbf{s})$ the probability that the next buyer met is type i in state $\mathbf{s}$

- Value of a match with a type-i buyer in state $\mathbf{s}$:

$$(\rho + \delta + \delta^B + \delta^S) V_{ji}^S(\mathbf{s}) = \tau_{ji}(\mathbf{s}) + \sigma_i^B(\mathbf{s})\, \theta^B \sum_{k=1}^{J} \upsilon_k^S \left[V_{ji}^S(\mathbf{s} + \mathbf{1}_k) - V_{ji}^S(\mathbf{s})\right]$$
$$+ (\delta + \delta^S) \sum_{k=1}^{J} (s_k - \mathbf{1}_{k=j}) \left[V_{ji}^S(\mathbf{s} - \mathbf{1}_k) - V_{ji}^S(\mathbf{s})\right]$$

- The match ends at rate $\delta + \delta^B + \delta^S$; the buyer adds a supplier at rate $\sigma_i^B \theta^B$; each of the buyer's *other* suppliers leaves at rate $\delta + \delta^S$

Bargaining over Transfers

- Each buyer and its suppliers divide the total surplus $V_i^T(\mathbf{s}) = V_i^B(\mathbf{s}) + \sum_j s_j V_{ji}^S(\mathbf{s})$. Transfers solve

$$\tau_{ji}(\mathbf{s}) = \frac{1}{2}\Big[\underbrace{\pi_i^T(\mathbf{s}) - \pi_i^T(\mathbf{s}-\mathbf{1}_j)}_{\text{supplier's effect on surplus}} - \underbrace{\sum_{k:s_k>0}(s_k - \mathbf{1}_{k=j})\big(\tau_{ki}(\mathbf{s}) - \tau_{ki}(\mathbf{s}-\mathbf{1}_j)\big)}_{\text{effect on transfers to other suppliers}}\Big]$$

- Assumptions behind it: (1) no commitment, so transfers are renegotiated whenever the buyer's supplier set changes; (2) simultaneous bilateral sessions run by one delegate per supplier (Binmore, Rubinstein and Wolinsky, 1986); (3) offers are private, breakdowns public; (4) passive beliefs; (5) contracts cannot condition on search effort or history
- Implications: (i) bilateral efficiency $\Rightarrow$ prices maximize π^T (de Fontenay and Gans); (ii) each pair splits its marginal surplus equally; (iii) the same transfers hold when bargaining over V^T rather than π^T
- Bargaining is one buyer to many suppliers, even though matching is many-to-many

Where the Transfer Rule Comes From

- Equal split of the bilateral flow surplus. The supplier's disagreement payoff is 0; the buyer's is its payoff with one fewer type-j supplier, with transfers to the others renegotiated:

$$\tau_{ji}(\mathbf{s}) = \Big[\pi_i^T(\mathbf{s}) - \sum_k s_k \tau_{ki}(\mathbf{s})\Big] - \Big[\pi_i^T(\mathbf{s}-\mathbf{1}_j) - \sum_k (s_k - \mathbf{1}_{k=j})\tau_{ki}(\mathbf{s}-\mathbf{1}_j)\Big]$$

- Split off the supplier's own transfer, $\sum_k s_k \tau_{ki}(\mathbf{s}) = \tau_{ji}(\mathbf{s}) + \sum_k (s_k - \mathbf{1}_{k=j})\tau_{ki}(\mathbf{s})$, and collect terms:

$$2\tau_{ji}(\mathbf{s}) = \pi_i^T(\mathbf{s}) - \pi_i^T(\mathbf{s}-\mathbf{1}_j) - \sum_k (s_k - \mathbf{1}_{k=j})\big[\tau_{ki}(\mathbf{s}) - \tau_{ki}(\mathbf{s}-\mathbf{1}_j)\big]$$

  which is the rule on the previous slide

- The system is recursive in $n^B(\mathbf{s})$: solve for portfolios with one supplier, then two, …
- An extra supplier helps the buyer twice: it adds retail surplus, and it lowers what the buyer pays its other suppliers (each becomes less critical at the margin)

Illustration: Homogeneous Suppliers

- One supplier type and $\pi^T(n) = A\, n^{\beta}$ with $\beta = \frac{\eta-1}{\alpha-1}$. The rule becomes $(n+1)\tau(n) = \Delta\pi(n) + (n-1)\tau(n-1)$
- Let $g(n) = n(n+1)\tau(n)$. Then $g(n) = n\,\Delta\pi(n) + g(n-1)$ with $g(0) = 0$, so

$$\tau(n) = \frac{1}{n(n+1)} \sum_{k=1}^{n} k \left[\pi^T(k) - \pi^T(k-1)\right]$$

  (the Stole–Zwiebel solution)

- Buyer's share of flow surplus, $1 - n\tau(n)/\pi^T(n)$:
  - $n = 1$: one half, whatever β. Linear surplus ($\alpha = \eta$, $\beta = 1$): $\tau = A/2$ for all n, and the buyer keeps one half
  - As $n \to \infty$: $\sum_k k\,\Delta\pi(k) \approx \frac{\beta}{1+\beta} A\, n^{1+\beta}$, so the buyer's share $\to \frac{1}{1+\beta}$
- At the estimates ($\eta = 2.432$, $\alpha = 4.35$): $\beta = 0.43$; the buyer's share rises from 0.50 ($n = 1$) to 0.65 ($n = 5$) and 0.69 ($n = 30$), with limit 0.70. Consistent with high-μ buyers keeping about two-thirds of gross profit

Illustration for this lecture; not in the paper.

Equations of Motion: Buyers

- $M_i^B(\mathbf{s})$: measure of type-i buyers in state $\mathbf{s}$. A buyer loses each supplier at rate $\tilde{\delta} \equiv \delta + \delta^S$. For $n^B(\mathbf{s}) > 0$:

$$\dot{M}_i^B(\mathbf{s}) = \sum_j \left[\mathbf{1}_{s_j > 0} \, \sigma_i^B(\mathbf{s} - \mathbf{1}_j) \theta^B \upsilon_j^S M_i^B(\mathbf{s} - \mathbf{1}_j) + \tilde{\delta} \, (s_j + 1) M_i^B(\mathbf{s} + \mathbf{1}_j) \right]$$
$$- \left[\sigma_i^B(\mathbf{s}) \theta^B + \tilde{\delta} \, n^B(\mathbf{s}) + \delta^B \right] M_i^B(\mathbf{s})$$

- Buyers with no suppliers (exited buyers rejoin this pool and keep searching):

$$\dot{M}_i^B(\mathbf{0}) = \delta^B \sum_{n^B(\mathbf{s}) \neq 0} M_i^B(\mathbf{s}) + \tilde{\delta} \sum_j M_i^B(\mathbf{1}_j) - \sigma_i^B(\mathbf{0}) \theta^B M_i^B(\mathbf{0})$$

- Inflows: a buyer in $\mathbf{s} - \mathbf{1}_j$ finds a type-j supplier; a buyer in $\mathbf{s} + \mathbf{1}_j$ loses one. Outflows: gain, loss, or exit
- Adding up: $\sum_{\mathbf{s}} M_i^B(\mathbf{s}) = M_i^B$, normalized so that $\sum_i M_i^B = 1$

Equations of Motion: Suppliers

- A supplier's search depends only on its number of buyers n^S; it loses each buyer at rate $\delta + \delta^B$ and exits at rate δ^S. For $n^S = 1, \ldots, N^S$:

$$\dot{M}_j^S(n^S) = \sigma_j^S(n^S - 1)\theta^S M_j^S(n^S - 1) + (\delta + \delta^B)(n^S + 1)M_j^S(n^S + 1) - \left[\sigma_j^S(n^S)\theta^S + (\delta + \delta^B)n^S + \delta^S\right] M_j^S(n^S)$$

- Suppliers with no buyers:

$$\dot{M}_j^S(0) = \delta^S \sum_{n^S \neq 0} M_j^S(n^S) + (\delta + \delta^B)\, M_j^S(1) - \sigma_j^S(0)\theta^S M_j^S(0)$$

- Adding up: $\sum_{n^S} M_j^S(n^S) = M_j^S$; the ratio of potential suppliers to potential buyers $M^S = \sum_j M_j^S$ is estimated
- Active suppliers per active buyer, $\sum_j (M_j^S - M_j^S(0)) / \sum_i (M_i^B - M_i^B(\mathbf{0}))$, is endogenous

The rate on $M_j^S(1)$ is $\delta + \delta^B$, which makes $\sum_n \dot{M}_j^S(n) = 0$. The zero-state supplier equation in the JPE manuscript has $\tilde{\delta} + \delta^B$, which double counts supplier exit.

Aggregates and Market Tightness

- Aggregate search effort on each side:

$$H^B = \sum_i \sum_{\mathbf{s}} \underbrace{\sigma_i^B(\mathbf{s}) M_i^B(\mathbf{s})}_{H_i^B(\mathbf{s})}, \qquad H^S = \sum_j \sum_{n^S} \underbrace{\sigma_j^S(n^S) M_j^S(n^S)}_{H_j^S(n^S)}$$

- Matches per unit time, from an urn–ball process (H^B urns, H^S balls):

$$X = H^B \left[1 - \left(1 - \tfrac{1}{H^B}\right)^{H^S}\right] \approx H^B \left[1 - e^{-H^S/H^B}\right]$$

  The approximation is homogeneous of degree one in (H^B, H^S)

- Match rates per unit of search: $\theta^B = X/H^B$, $\theta^S = X/H^S$; both are functions of tightness H^S/H^B
- Composition of the other side of the market:

$$v_i^B(\mathbf{s}) = \frac{H_i^B(\mathbf{s})}{H^B}, \qquad v_j^S = \frac{\sum_{n^S} H_j^S(n^S)}{H^S}$$

Steady-State Equilibrium

Given potential masses $\{M_i^B\}$, $\{M_j^S\}$ and spending E, a steady state consists of retail prices, transfers $\tau_{ji}(\mathbf{s})$, search efforts $\sigma_i^B(\mathbf{s})$ and $\sigma_j^S(n^S)$, match rates (θ^B, θ^S), and distributions $M_i^B(\mathbf{s})$, $M_j^S(n^S)$ such that

1. Consumers maximize utility, buyers charge the markup $\frac{\eta}{\eta-1}$, and the product market clears
2. Buyers and suppliers solve their HJB equations and search problems; transfers satisfy the bargaining rule
3. Distributions are stationary: $\dot{M}_i^B(\mathbf{s}) = 0$, $\dot{M}_j^S(n^S) = 0$, a system of $I(\|\mathbb{S}\| + 1) + J(N^S + 1)$ equations
4. θ^B and θ^S are consistent with aggregate search through the matching function

- Transition paths after unexpected shocks are also computed (needed for the tariff experiment)
- The equilibrium need not be efficient: the usual congestion externalities, plus buyers' incentive to over-accumulate suppliers to lower transfers

Estimation and Baseline Equilibrium

Estimation Strategy

- Calibrated: within-buyer elasticity $\alpha = 4.35$ (Hottman, Redding and Weinstein, 2016); buyer exit $\delta^B = 0.07$ (U.S. retail exit rate); supplier exit $\delta^S = 0.15$ (Chinese apparel producers, 2004–06)
- Estimated without solving the model: the overall match death rate of 0.774 (Poisson fit) implies $\delta = 0.774 - 0.07 - 0.15 = 0.554$
- GMM for $\Lambda = (k_0, \gamma^B, \gamma^S, M^S, \Delta, \omega, \sigma^2_{\ln \mu}, \eta)$:

$$\hat{\Lambda} = \arg\min_{\Lambda} \; \big(m(\Lambda) - \bar{m}\big)\, W\, \big(m(\Lambda) - \bar{m}\big)'$$

  with 209 moments from Tables 1–4 plus the variable cost–revenue ratio of U.S. apparel retailers; W is block diagonal
- Types: $\ln \mu \sim$ discretized $N(0, \sigma^2_{\ln \mu})$; $\ln \tilde{c}$ has two mass points, Δ apart, with share ω at the low-cost (high-quality) point
- Identifying η: summing the markup rule $q_{xy} p_{xy} \frac{\eta - 1}{\eta} = q_{xy} c_x$ over all matches gives $\frac{\eta - 1}{\eta} = \frac{C}{R}$, total variable cost over total retail revenue (in practice adjusted for transfers to suppliers)

Identification of the Other Parameters

- The mapping from moments to parameters is many-to-many (Andrews, Gentzkow and Shapiro, 2017 sensitivity), but
- **Transition matrices and degree distributions** (Tables 1–3) $\rightarrow k_0, \gamma^B, \gamma^S$, which govern the rates at which firms in each state add partners, and M^S, which governs expected connections on each side
- **Payments per supplier vs. number of suppliers** (Table 4, column 2) $\rightarrow \sigma^2_{\ln \mu}$: with random matching, portfolio composition is similar across buyers, so spending per supplier mainly reflects buyer type
- **Within-buyer supplier shares** (Table 4, columns 3–12) $\rightarrow \Delta, \omega$: with one supplier type, each supplier's share would be $1/n$; dispersion in shares reveals dispersion in $\tilde{c}$

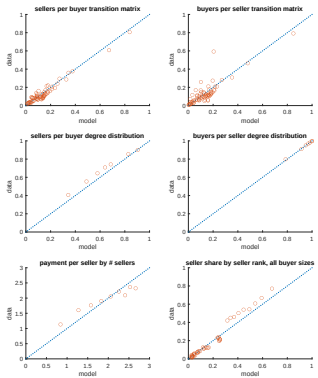
Parameter Estimates

	Estimate	S.E.
k_0	0.009	0.003
γ^B	0.320	0.041
γ^S	0.230	0.046
ω	0.030	0.002
Δ	0.454	0.006
M^S	4.203	0.728
η	2.432	0.141
$\sigma^2_{\ln \mu}$	7.428	2.508
Objective	10,461.73	

EJTX, Table 5.

- Visibility effects on both sides, stronger for buyers ($\gamma^B > \gamma^S$): buyers face diminishing returns and large buyers must replace many expiring matches
- For a given σ, a buyer with 30 suppliers pays $(2/31)^{0.32} \approx 0.42$ of the search cost of a buyer with one, i.e. 58% less
- Large spread in buyer appeal: fat right tail of suppliers per buyer
- $M^S = 4.2$ potential suppliers per potential buyer, but 8.48 *active* suppliers per active buyer (2.88 excluding single-match firms): many suppliers serve the U.S. incidentally
- Only 3% of potential suppliers are high quality, about 45% more appealing per dollar

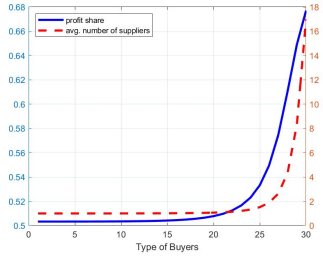
Model Fit



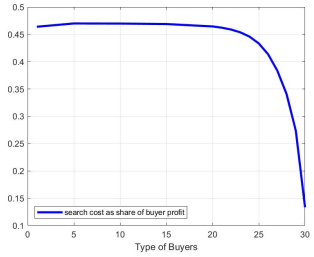
EJTX, Figure 8.

- Fits the 140 targeted transition probabilities well; only one is substantially off (large suppliers staying large)
- Fits the steady-state degree distributions; slightly underpredicts buyers with many suppliers
- Reproduces the rise in spending per supplier with the number of suppliers, but flatter (only two supplier types)
- Variable cost–revenue ratio: 0.731 in the data, 0.730 in the model

Profits, Bargaining Power and Search Costs



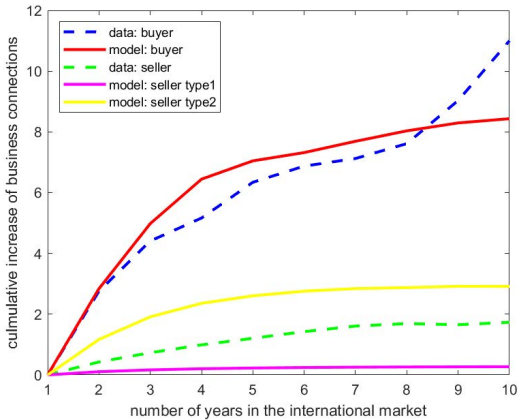
Buyer share of gross profit



Buyer search cost as share of flow profit

- Buyers and suppliers jointly keep $1/\eta = 0.41$ of spending as gross profit. Low-μ buyers have one supplier and split 50–50; high-μ buyers have many suppliers and keep about two-thirds
- High-quality suppliers are 3% of potential but 12% of active suppliers, and earn 0.11 of spending vs. 0.03 for low-quality suppliers
- Buyers spend 16% of gross profit on search (about 45% for low-μ, under 10% for high-μ); suppliers spend 0.064 of spending, about 45% of their profit

Life Cycle of Buyers and Suppliers



EJTX, Figure 10. Data from LFTTD apparel shipment records.

- Not targeted in estimation
- Buyers active for 10 years have about nine more suppliers than first-year buyers; the model matches this through rapid accumulation by high-μ buyers and their overrepresentation in older cohorts
- Suppliers gain only about 1.5 buyers over 10 years; the model underpredicts this growth by about half, since most suppliers are low type
- Visibility speeds accumulation; convex costs of replacing expiring matches eventually stop it

Experiments

Interpreting Market Developments, 1996–2011

- Three patterns to explain
  - Number of suppliers jumps 28% in 2005, while exports per supplier fall about 15% (Chinese entry: ATC phaseout and the 2004 removal of restrictions on direct exporting by small private firms)
  - Suppliers rise about 25% between 1996 and 2004 with flat consumption (interpreted as an "IT shock" to search costs)
- Three steady states
  - 2006–2011: the baseline estimates
  - 2004 (post-IT, pre-policy): choose (M^S, ω) to match growth in active exporters and sales per exporter after 2004. Result: ω falls from 0.043 to 0.030 and M^S rises from 2.4 to 4.2
  - 1996 (pre-IT, pre-policy): raise k_0 to match the smaller number of active exporters, which requires $k_0^{1996} = 1.65 \times k_0^{2004}$

Counterfactual: Interpreting Market Developments

		(1) 1996 low M^S, high k_0	(2) 2004 low M^S, base k_0	(3) 2011 base M^S, k_0
1.	measure, active low-quality suppliers	0.480	21.6%	57.2%
2.	measure, active high-quality suppliers	0.087	4.2%	24.3%
3.	measure, active buyers	0.190	16.7%	15.6%
4.	total profit, low-quality suppliers	0.024	-0.3%	24.5%
5.	total profit, high-quality suppliers	0.117	-1.4%	-7.8%
6.	total profit, buyers	0.270	0.6%	1.2%
7.	total search costs, low-quality suppliers	0.010	0.1%	24.2%
8.	total search costs, high-quality suppliers	0.056	4.0%	-4.9%
9.	total search costs, buyers	0.046	-14.8%	-4.8%
10.	number of suppliers per buyer	1.695	11.3%	17.9%
11.	high-quality suppliers per buyer	0.581	11.7%	-6.8%
12.	consumer welfare	1.000	8.0%	7.1%

EJTX, Table 6. Column (1) in levels (measures relative to potential buyers; profits and costs relative to spending; welfare normalized to 1). Columns (2)–(3): percent change from column (1).

Lessons: Search Costs vs. Market Access

- **IT shock** (column 2): welfare $+8\%$, from 17% more retailers and 11% more suppliers per buyer. More supplier competition lets buyers cut search and keep slightly more profit

- **Policy shock** (column 3 vs. 2): a congestion effect
  - The surge of low-quality suppliers makes it harder for high-quality suppliers to match, so they search less
  - Suppliers per buyer rise, but high-quality suppliers per buyer fall, below even the 1996 level
  - Weaker search incentives push some low-μ buyers out, shifting shares toward “big box” stores
  - Welfare falls relative to column 2 (7.1% vs. 8.0%)

- Contrast: the canonical sunk-cost model of exporting (Alessandria, Arkolakis and Ruhl, 2021), calibrated to the same experiment, predicts a large welfare gain from love of variety. Two-sided search and congestion can flip the sign

Section 301 Tariffs on Chinese Apparel

- Chinese suppliers treated as low quality (lower value per exporter); unexpected, permanent 15 percentage point tariff
- On impact: only intensive margins move. For a buyer with s_1 low- and s_2 high-quality suppliers, the high-quality share is

$$h_{2|i} = \frac{s_2 \tilde{c}_2^{1-\alpha}}{s_1\big((1+t_1)\tilde{c}_1\big)^{1-\alpha} + s_2 \tilde{c}_2^{1-\alpha}}$$

- Over time: search efforts adjust, matches die off, the network rewires

	Long-run change
active low-quality suppliers	-22.4%
active high-quality suppliers	3.4%
active buyers	1.6%
profit, low-quality suppliers	-52.5%
profit, high-quality suppliers	18.3%
profit, buyers	-2.2%
suppliers per buyer	-7.6%
high-quality suppliers per buyer	28.1%
consumer welfare	-1.0%

EJTX, Table 7.

Transition Dynamics: Where Do They Come From?

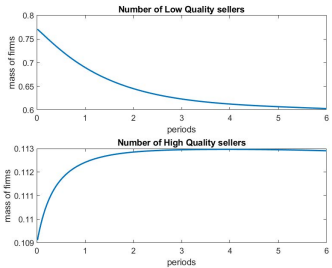
- Share of consumer spending on type-j goods along the transition:

$$R_{jt} = \sum_{\mathbf{s}} s_j \underbrace{\left(\frac{(1+t_j)\tilde{c}_j}{\tilde{c}(\mathbf{s})} \right)^{1-\alpha}}_{\text{intensive margin within buyer}} \underbrace{\frac{\sum_i M_{it}^B(\mathbf{s})\, \mu_i^{\eta-1}\, \tilde{c}(\mathbf{s})^{1-\eta}}{\sum_{\mathbf{s}'} \sum_i M_{it}^B(\mathbf{s}')\, \mu_i^{\eta-1}\, \tilde{c}(\mathbf{s}')^{1-\eta}}}_{\text{extensive margin: buyers' supplier composition}}$$

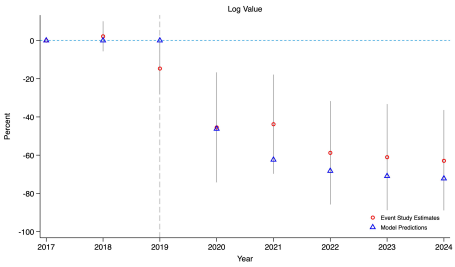
  with $\tilde{c}(\mathbf{s}) = \left[\sum_k s_k \big((1+t_k)\tilde{c}_k\big)^{1-\alpha} \right]^{\frac{1}{1-\alpha}}$

- Derivation: a buyer's share of E is proportional to $\mu_i^{\eta-1} P_y^{1-\eta} \propto \mu_i^{\eta-1} \tilde{c}(\mathbf{s})^{1-\eta}$; within the buyer, type j gets share $s_j \big((1+t_j)\tilde{c}_j / \tilde{c}(\mathbf{s})\big)^{1-\alpha}$
- The only source of dynamics is the endogenous evolution of buyers' supplier portfolios $M_{it}^B(\mathbf{s})$, driven by search on both sides
- Payments to type-j exporters: variable-cost compensation $\frac{\eta-1}{\eta} R_j$ plus bargained transfers; used to compare with the data

Tariff Transition: Model and Event-Study Evidence



Transition of supplier masses



China vs. other exporters: data and model

- High-quality suppliers search harder immediately: their mass reaches the new steady state within about 3 years; low-quality relationships dissolve slowly, over about 6 years
- Event study, $\ln y_{jgt} = \alpha_{jg} + \alpha_{gt} + \sum_{\tau} \beta_{\tau} \mathbf{1}[t - 2019 = \tau] \times target_{jg} + \varepsilon_{jgt}$, with Vietnam, Bangladesh, Indonesia, India and Mexico as controls: the 2020 drop is intensive margin; the continued decline is extensive-margin reallocation

Welfare Dynamics and Takeaways

- Consumer welfare falls 2.2% when the tariff is imposed (pass-through to retail prices)
- High-quality suppliers quickly gain share, so welfare rebounds and slightly overshoots
- Gradual exit of low-quality suppliers brings welfare to a steady-state loss of about 1%

Takeaways from EJTX

- Search costs are borne about equally by buyers and suppliers in aggregate; buyers pay less per match and keep more of the surplus
- Conditional on survival, a successful buyer takes about 10 years to reach its long-run size, a supplier about 5: search frictions generate long lags in aggregate trade responses
- Lower search costs raise welfare but shift rents away from firms; more market access need not raise welfare because of congestion
- Policies that hit a *subset* of suppliers interact with congestion: the 2018–19 tariffs were partly offset by reallocation toward higher-quality suppliers

Where Did and Do We Go from Here?

- Micro level
  - More detailed firm-to-firm data: domestic and international links together
  - What is “demand” ξ_t? Sales per customer vs. number of customers, and hence the nature of trade costs
  - Long-term relationships and contracting: closer links to the global value chain literature
  - Directed rather than random search; combinatorial choices of destinations and suppliers
- Aggregate level
  - Policy uncertainty and trade wars: Handley and Limão (2017); Alessandria et al. (2025)
  - Short-run vs. long-run trade elasticities and the welfare cost of transitions
  - More realistic policy: regional agreements, partial coverage, export controls

Summary

- Export status is persistent; entrants start small and grow; entry matters for aggregates only slowly
- Sunk costs generate hysteresis: the band between entry and exit thresholds is the annuitized sunk cost, widened by uncertainty
- The canonical model misses post-entry dynamics. Customer capital (marketing) fits quantity growth with flat prices; R&D links exporting to productivity growth
- In general equilibrium, long-run trade elasticities are far larger than short-run ones, and steady-state comparisons misstate welfare
- Expectations matter: much of U.S.–China trade growth after 2001 reflects reduced uncertainty in 1986–93 and slow adjustment to the 1980 reform
- Two-sided search: match-level dynamics, bargaining power, and congestion. More market access can lower welfare; tariffs on a subset of suppliers trigger reallocation

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Appendix: Continuous-Time Bellman Equations

Stochastic Bellman Equations

- Consider the value function

$$v(x(t), t) = \max_u \mathbb{E}\Big[\int_t^\infty e^{-\rho(s-t)} \pi(x(s), u(s))\, ds \,\Big|\, x(t) \Big], \qquad t \geq 0$$

  where $x(t)$ follows a stochastic process

- What is the Bellman equation for this problem?
- Discrete-time analogue with period length $\Delta t > 0$, $t' = t + \Delta t$, $x' = x + \Delta x$:

$$v(x, t) = \max_u \; \pi(x, u)\Delta t + \frac{1}{1 + \rho \Delta t} \mathbb{E}[v(x', t') \mid x]$$

Stochastic Bellman Equation

- Multiply both sides by $(1 + \rho\Delta t)$ and subtract $v(x, t)$:

$$\rho\, v(x,t)\Delta t = \max_u\ \pi(x,u)\Delta t(1+\rho\Delta t) + \mathbb{E}[v(x',t') - v(x,t)\,|\,x]$$

- Write out the change in the value function:

$$v(x',t') - v(x,t) = \big[v(x',t') - v(x,t')\big] + \frac{v(x,t') - v(x,t)}{\Delta t}\,\Delta t$$

- The second change is *deterministic* and can be pulled outside the expectation:

$$\rho\, v(x,t)\Delta t = \max_u \Big\{\pi(x,u)\Delta t(1+\rho\Delta t) + \mathbb{E}\big[v(x',t') - v(x,t')\,|\,x\big] + \frac{v(x,t') - v(x,t)}{\Delta t}\,\Delta t\Big\}$$

- How to treat $\mathbb{E}[v(x',t') - v(x,t')\,|\,x]$ depends on the stochastic process for x. The Poisson case is especially convenient in continuous time because x moves in discrete steps

Poisson Process

- Continuous-time process $x(t)$ with $x(0) = 0$ and
  - (i) stationary, independent increments
  - (ii) increments with a Poisson distribution with intensity $\lambda > 0$
- The increment $\Delta x = x(t + \Delta t) - x(t)$ satisfies

$$\Pr[\Delta x = k] = \frac{(\lambda \Delta t)^k e^{-\lambda \Delta t}}{k!}$$

  which depends only on Δt, not on t
- In particular $\Pr[x(t) = k] = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$, so $\mathbb{E}[x(t)] = \mathsf{Var}[x(t)] = \lambda t$
- Sample paths are discontinuous: a counting process

Properties of the Poisson Process

- Over an interval of length Δt:

$$\Pr[\Delta x = 1] = (\lambda \Delta t) e^{-\lambda \Delta t}, \qquad \Pr[\Delta x = 0] = e^{-\lambda \Delta t}$$

  (note the link to the exponential distribution of waiting times)

- Expanding,

$$\begin{aligned} \Pr[\Delta x = 1] &= \lambda \Delta t + o(\Delta t) \\ \Pr[\Delta x = 0] &= 1 - \lambda \Delta t + o(\Delta t) \\ \Pr[\Delta x > 1] &= o(\Delta t) \end{aligned}$$

  where $o(\Delta t)/\Delta t \to 0$ as $\Delta t \to 0$

Bellman Equation When the State Is Poisson

- Write out the expectation:

$$\begin{aligned}\mathbb{E}[v(x',t') - v(x,t') \mid x] &= \sum_{k=0}^{\infty} \big[v(x+k,t') - v(x,t')\big] \Pr[\Delta x = k] \\ &= \big[v(x+1,t+\Delta t) - v(x,t+\Delta t)\big]\lambda\Delta t + o(\Delta t)\end{aligned}$$

  since the $k = 0$ term is zero and the terms with $k \geq 2$ are $o(\Delta t)$

- Plug into the Bellman equation:

$$\begin{aligned}\rho\, v(x,t)\Delta t = \pi(x)\Delta t(1+\rho\Delta t) &+ \big[v(x+1,t+\Delta t) - v(x,t+\Delta t)\big]\lambda\Delta t \\ &+ \frac{v(x,t') - v(x,t)}{\Delta t}\Delta t + o(\Delta t)\end{aligned}$$

- Divide by Δt and let $\Delta t \to 0$:

$$\rho\, v(x,t) = \pi(x) + \lambda\big(v(x+1,t) - v(x,t)\big) + \frac{\partial v}{\partial t}$$

Birth–Death Processes

- Generalize to competing events: the waiting time to the next "birth" is exponential with rate λ, to the next "death" exponential with rate μ
- The next event arrives at rate $\lambda + \mu$; conditional on an event, it is a birth with probability $\frac{\lambda}{\lambda+\mu}$ and a death with probability $\frac{\mu}{\lambda+\mu}$

$$\Pr[\Delta x = 1] = \lambda \Delta t + o(\Delta t), \qquad \Pr[\Delta x = -1] = \mu \Delta t + o(\Delta t)$$
$$\Pr[\Delta x = 0] = 1 - (\lambda + \mu)\Delta t + o(\Delta t), \qquad \Pr[|\Delta x| > 1] = o(\Delta t)$$

- The same steps give

$$\rho\, v(x,t) = \pi(x) + \lambda\big(v(x+1,t) - v(x,t)\big) + \mu\big(v(x-1,t) - v(x,t)\big) + \frac{\partial v}{\partial t}$$

- This is the structure of the EJTX HJB equations: partners arrive at rate $\sigma\theta v_j$ and leave at rate $s_j(\delta + \delta^S)$, with the state a vector of counts